ASSIGNMENT 4

Deadline: 11:59 pm, March 29, 2026

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1 Compulsory Questions

Question 1 (20%)

Let the nonlinear feature transform be defined as:

$$\mathbf{z} = \phi_2(x_1, x_2) = (1, x_1, x_2, x_1^2, x_1x_2, x_2^2) \in \mathcal{Z}.$$

Please use a hyperplane in $\mathcal{Z}$ to represent the following boundaries in the original space (x_1, x_2).

1. **Represent the ellipse:** $2(x_1 - 3)^2 + (x_2 - 4)^2 = 1$.
2. **Represent the hyperbola:** $(x_1 - 3)^2 - (x_2 - 4)^2 = 1$..

Question 2(10%)

If $\lambda < 0$ in the augmented error

$$E_{\text{aug}}(\mathbf{w}) = E_{\text{in}}(\mathbf{w}) + \lambda \mathbf{w}^T \mathbf{w},$$

what soft order constraint does this correspond to?

Hint: $\lambda < 0$ encourages large weights.

Question 3(20%)

Given $E_{aug}(\mathbf{w}) = E_{in}(\mathbf{w}) + \lambda \mathbf{w}^\top \mathbf{w}$, $\mathbf{w}_{reg} = \arg\min_{\mathbf{w}} E_{aug}(\mathbf{w})$ and $\mathbf{w}_{lin} = \arg\min_{\mathbf{w}} E_{in}(\mathbf{w})$, please show that the norm of the regularized weight vector satisfies the following inequality:

$$\|\mathbf{w}_{reg}\| \leq \|\mathbf{w}_{lin}\|.$$

Hint: Begin by assuming the opposite, i.e., $\|\mathbf{w}_{reg}\| > \|\mathbf{w}_{lin}\|$, and derive a contradiction.

Question 4 (20%)

Given that E_{in} is differentiable, we apply gradient descent to minimize the augmented error $E_{aug}(\mathbf{w}) = E_{in}(\mathbf{w}) + \lambda \mathbf{w}^\top \mathbf{w}$:

$$\mathbf{w}^{(t+1)} \leftarrow \mathbf{w}^{(t)} - \eta \nabla_{\mathbf{w}} E_{aug}(\mathbf{w}^{(t)})$$

where t denotes the index of iterations.

Please show that the update rule above is equivalent to:

$$\mathbf{w}^{(t+1)} \leftarrow (1 - 2\eta\lambda)\mathbf{w}^{(t)} - \eta \nabla_{\mathbf{w}} E_{in}(\mathbf{w}^{(t)}).$$

Hint: Expand $E_{aug}(\mathbf{w}^{(t)})$ and interpret the resulting update.

Note: This explains the term "weight decay": $\mathbf{w}^{(t)}$ decays before being updated by the gradient of E_{in}.

Question 5 (30%)

Programming question. In this problem, you will implement logistic regression and study the effect of ℓ_2 regularization. The data is included in `q5_x_train.csv`, `q5_y_train.csv`, `q5_x_test.csv`, and `q5_y_test.csv`; please use them directly.

Given a binary classification dataset

$$\{(x^{(i)}, y^{(i)})\}_{i=1}^m, \qquad x^{(i)} \in \mathbb{R}^d, \quad y^{(i)} \in \{-1, 1\}.$$

We augment each input with a bias term and write

$$\tilde{x}^{(i)} = \begin{bmatrix} 1 \\ x_1^{(i)} \\ \vdots \\ x_d^{(i)} \end{bmatrix}, \qquad w = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \end{bmatrix}.$$

For logistic regression, we optimize the empirical loss

$$L(w) = \frac{1}{m} \sum_{i=1}^{m} \log\Big(1 + e^{-y^{(i)} w^T \tilde{x}^{(i)}}\Big).$$

Part 1: Logistic Regression without Regularization
The gradient of the loss with respect to w is

$$\nabla L(w) = -\frac{1}{m} \sum_{i=1}^{m} \frac{y^{(i)} \tilde{x}^{(i)}}{1 + e^{y^{(i)} w^T \tilde{x}^{(i)}}}.$$

The gradient descent update is

$$w^{(t+1)} \leftarrow w^{(t)} - \eta \nabla L(w^{(t)}).$$

1. Prepare the input matrix by augmenting each data point with a bias term 1.
2. Initialize
$$w = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^{d+1}.$$
3. Use gradient descent to train logistic regression on the training set (`q5_x_train.csv`, `q5_y_train.csv`). Use learning rate $\eta = 0.01$. Stop the algorithm when the Euclidean norm of the gradient is smaller than 10^{-2}.
4. Plot the training loss curve. [Hint: Record the value of the loss function at each iteration.]
5. After training, report the final training accuracy, the final test accuracy on the test set (`q5_x_test.csv`, and `q5_y_test.csv`), and the norm of the non-bias weights
$$\|w_{1:d}\|_2 = \sqrt{\sum_{j=1}^{d} w_j^2}.$$

Part 2: Logistic Regression with ℓ_2 Regularization
Now consider ℓ_2-regularized logistic regression:

$$L_\lambda(w) = L(w) + \frac{\lambda}{2m} \sum_{j=1}^{d} w_j^2,$$

where the bias term w_0 is not regularized. Therefore,

$$L_\lambda(w) = \frac{1}{m} \sum_{i=1}^{m} \log\Big(1 + e^{-y^{(i)} w^T \tilde{x}^{(i)}}\Big) + \frac{\lambda}{2m} \sum_{j=1}^{d} w_j^2.$$

The gradient of the regularized objective with respect to w_0 is

$$\frac{\partial L_\lambda(w)}{\partial w_0} = -\frac{1}{m} \sum_{i=1}^{m} \frac{y^{(i)} \tilde{x}_0^{(i)}}{1 + e^{y^{(i)} w^T \tilde{x}^{(i)}}},$$

and for $j = 1, \ldots, d$,

$$\frac{\partial L_\lambda(w)}{\partial w_j} = -\frac{1}{m} \sum_{i=1}^{m} \frac{y^{(i)} \tilde{x}_j^{(i)}}{1 + e^{y^{(i)} w^T \tilde{x}^{(i)}}} + \frac{\lambda}{m} w_j.$$

Set $\lambda = 1$.

1. Starting again from the zero weight vector, train the regularized logistic regression model on the training set. Use learning rate $\eta = 0.01$. Stop the algorithm when the Euclidean norm of the gradient is smaller than 10^{-2}.
2. Plot the training loss curve.
3. After training, report the final training accuracy, the final test accuracy, and the norm of the non-bias weights $\|w_{1:d}\|_2$.
4. Compare the training accuracy, test accuracy, and weight norm with and without ℓ_2 regularization.

Question 1

$\tilde{w}^T \tilde{z} = 0$

$w_0 + w_1 x_1 + w_2 x_2 + w_3 x_1^2 + w_4 x_1 x_2 + w_5 x_2^2 = 0$

1. $2(x_1 - 3)^2 + (x_2 - 4)^2 = 1$

$2x_1^2 - 12x_1 + 18 + x_2^2 - 8x_2 + 15 = 0$

$33 - 12x_1 - 8x_2 + 2x_1^2 + x_2^2 = 0$

By comparing like terms,

$w_0 = 33$

$w_1 = 12$

$w_2 = -8$

$w_3 = 2$

$w_4 = 0$

$w_5 = 1$

$$\therefore \tilde{w}_{ellipse} = \begin{bmatrix} 33 \\ -12 \\ -8 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

2) $(x_1 - 3)^2 - (x_2 - 4)^2 = 1$

$x_1^2 - 6x_1 + 9 - x_2^2 + 8x_2 - 17 = 0$

$-8 - 6x_1 + 8x_2 + x_1^2 - x_2^2 = 0$

$$\therefore \tilde{w}_{hyperbola} = \begin{bmatrix} -8 \\ -6 \\ 8 \\ 1 \\ 0 \\ -1 \end{bmatrix}$$

Question 2

When $\lambda < 0$:

Let $\alpha = -\lambda > 0$.

The augmented error becomes $E_{aug}(\vec{w}) = E_{in}(\vec{w}) + \lambda \vec{w}^T \vec{w}$

$$= E_{in}(\vec{w}) - \alpha \vec{w}^T \vec{w}$$

In Lagrangian framework, minimize $E_{in}(\vec{w})$ to inequality constaint $g(\vec{w}) \leq 0$.

$$L(\vec{w}, \alpha) = E_{in}(\vec{w}) + \alpha g(\vec{w}), \quad \alpha \geq 0.$$

$$g(\vec{w}) = -\vec{w}^T w + C \leq 0$$

$$\vec{w}^T w \geq C$$

$$\|\vec{w}\|^2 \geq C$$

where C is a positive constraint.

$$\therefore E_{aug}(\vec{w}) = E_{in}(\vec{w}) - \alpha \|\vec{w}\|^2$$

Question 3

Prove $\|\vec{w}_{reg}\| \le \|\vec{w}_{lin}\|$

Let $\vec{w}_{lin} = \arg\min_{\vec{w}} E_{in}(\vec{w})$, $E_{in}(\vec{w}_{lin}) \le E_{in}(\vec{w}_{reg})$

Let $\vec{w}_{reg} = \arg\min_{\vec{w}} E_{aug}(\vec{w})$, $E_{aug}(\vec{w}) = E_{in}(\vec{w}) + \lambda \vec{w}^T \vec{w}$

Assume $\|\vec{w}_{reg}\| > \|\vec{w}_{lin}\|$

$$E_{in}(\vec{w}_{reg}) + \lambda\|\vec{w}_{reg}\|^2 \le E_{in}(\vec{w}_{lin}) + \lambda\|\vec{w}_{lin}\|^2$$

$$E_{in}(\vec{w}_{reg}) - E_{in}(\vec{w}_{lin}) \le \lambda\left(\|\vec{w}_{lin}\|^2 - \|\vec{w}_{reg}\|^2\right)$$

$\because$ $\vec{w}_{lin}$ minimizes E_{in},

$\therefore$ $$0 \le E_{in}(\vec{w}_{reg}) - E_{in}(\vec{w}_{lin}) \le \lambda\left(\|\vec{w}_{lin}\|^2 - \|\vec{w}_{reg}\|^2\right)$$

$\because$ $\lambda \ge 0$,

$\therefore$ $\|\vec{w}_{lin}\|^2 - \|\vec{w}_{reg}\|^2 \ge 0$

$$\|\vec{w}_{lin}\| \ge \|\vec{w}_{reg}\|$$

$\because$ It contradicts our assumption

$\therefore$ $\|w_{reg}\| \le \|w_{lin}\|$

Question 4.

$$E_{aug}(\vec{w}) = E_{in}(\vec{w}) + \lambda \vec{w}^T \vec{w}$$

$$\nabla_{\vec{w}} E_{aug}(\vec{w}) = \nabla_{\vec{w}} E_{in}(\vec{w}) + \nabla_{\vec{w}}(\lambda \vec{w}^T \vec{w})$$

$$\nabla_{\vec{w}} E_{aug}(\vec{w}) = \nabla_{\vec{w}} E_{in}(\vec{w}) + 2\lambda \vec{w}$$

Applying gradient descent,

$$\vec{w}^{(t+1)} \leftarrow \vec{w}^{(t)} - \eta \nabla_{\vec{w}} E_{aug}(\vec{w}^{(t)})$$

$$\vec{w}^{(t+1)} \leftarrow \vec{w}^{(t)} - \eta \left(\nabla_{\vec{w}} E_{in}(\vec{w}^{(t)}) + 2\lambda \vec{w}^{(t)} \right)$$

$$\vec{w}^{(t+1)} \leftarrow \vec{w}^{(t)} - 2\eta\lambda \vec{w}^{(t)} - \eta \nabla_{\vec{w}} E_{in}(\vec{w}^{(t)})$$

$$\vec{w}^{(t+1)} \leftarrow (1 - 2\eta\lambda) \vec{w}^{(t)} - \eta \nabla_{\vec{w}} E_{in}(\vec{w}^{(t)})$$

Question 5

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PART 1 (lambda=0):
  - Training Accuracy: 98.74%
  - Test Accuracy:     97.66%
  - Weight Norm ||w||: 2.9646
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PART 2 (lambda=1):
  - Training Accuracy: 98.49%
  - Test Accuracy:     98.25%
  - Weight Norm ||w||: 2.5124
==================================================
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Comparison:

After applying l_2 regularization with $\lambda = 1$, $\|w_{1:d}\|_2$ decreases from 2.9646 to 2.5124. It shows that regularization constrains the magnitude of the weights. Though training accuracy slightly drops, test accuracy improves. It demonstrates that l_2 regularization reduces overfitting. The model sacrifices small amount of training performance to obtain a smoother decision boundary. It leads to better generalization on unseen test data.

Attachment

Declaration for written assignment

I am/we are submitting the assignment for:
☑ an individual project or
□ a group project on behalf of all members of the group. It is hereby confirmed that the submission is authorized by all members of the group, and all members of the group are required to sign this declaration.

I/We declare that: (i) the assignment here submitted is original except for source material explicitly acknowledged/all members of the group have read and checked that all parts of the piece of work, irrespective of whether they are contributed by individual members or all members as a group, here submitted are original except for source material explicitly acknowledged; (ii) the piece of work, or a part of the piece of work has not been submitted for more than one purpose (e.g. to satisfy the requirements in two different courses) without declaration; and (iii) the submitted soft copy with details listed in the <Submission Details> is identical to the hard copy(ies), if any, which has(have) been / is(are) going to be submitted. I/We also acknowledge that I am/we are aware of the University's policy and regulations on honesty in academic work, and of the disciplinary guidelines and procedures applicable to breaches of such policy and regulations, as contained in the University website http://www.cuhk.edu.hk/policy/academichonesty/.

In the case of a group project, we are aware that all members of the group should be held responsible and liable to disciplinary actions, irrespective of whether he/she has signed the declaration and whether he/she has contributed, directly or indirectly, to the problematic content.

I/we declare that I/we have not distributed/ shared/ copied any teaching materials without the consent of the course teacher(s) to gain unfair academic advantage in the assignment/ course.

I/we declare that I/we have read and understood the University's policy on the use of AI for academic work. I/we confirm that I/we have complied with the instructions given by my/our course teacher(s) regarding the use of AI tools for this assignment and consent to the use of AI content detection software to review my/our submission.

I/We also understand that assignments without a properly signed declaration by the student concerned and in the case of a group project, by all members of the group concerned, will not be graded by the teacher(s).

Chena	29 / 3 / 2025
Signature(s)	Date
Fung Cheuk Nam	1155192095
Name~~(s)~~	Student ID(s)
CSCI 3320	Fundamental of Machine Learning
Course code	Course title

Use of AI tools.

I acknowledge the use of Grok, Gemini (2026, Mar 29)

for content guidance, code guidance